

# Option Pricing, Computational Finance and Machine Learning

## Lesson 2

### Arbitrage Theory in Continuous Time

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# A Theoretical Approach to Derivatives Pricing

- Lesson 2 generalizes and formalizes some results seen in lesson 1, and reviews the Modern Theory of Asset Pricing, mainly following Bjork's book *Arbitrage Theory in Continuous Time*, 4th ed.
- Objectives are:
  - ▶ Consistently model the price series of *multiple* underlying assets
  - ▶ Build theoretical grounds for pricing and hedging *arbitrary* payoffs, not just European calls
  - ▶ Support rigorous modeling of interest rates, under both deterministic or stochastic assumptions, thereby building theoretical foundations for lesson 3 and beyond
- We skip the obligatory introduction of binomial models, leaving them for a programming assignment

# Outline

- 1 Assumptions
- 2 Girsanov's Theorem
- 3 Fundamental Asset Pricing Theorems
- 4 Numeraires

# Assumptions

# Idealized Markets

- No transaction costs, infinite liquidity, transparent and costless short-selling
  - No dividends
  - No rates
- for now:** we include support for rates later in this lesson

# Asset Price Series

- Finite horizon  $T > 0$  where everything is resolved, so all random variables are  $T$ -measurable
- Filtration generated by a  $p$ -dimensional standard Brownian Motion, a vector  $(W_t) \in \mathbb{R}^p$ , of  $p$  independent Brownian Motions – this means that  $(W_t)$  is the only source of randomness: all  $t$ -measurable (square-integrable) variables are deterministic functionals of Brownian paths  $\{W_s, 0 \leq s \leq t\}$
- Universe of  $n$  assets, which price series  $(S_t) \in \mathbb{R}^n$  are semi-martingales – i.e. it exists two adapted processes  $(\mu_t) \in \mathbb{R}^n$  and  $(\sigma_t) \in \mathbb{R}^{n \times p}$  such that:

$$dS_t = \mu_t dt + \sigma_t dW_t$$

- Essentially translates to assuming continuous paths with technical regularity, see lesson 1
- Assume  $(\mu_t)$  and  $(\sigma_t)$  are square-integrable:

$$\int_0^T \mu'_t \mu_t dt < \infty \text{ and } \int_0^T \text{tr}(\sigma'_t \sigma_t) dt < \infty \text{ where ' denotes transposition}$$

- Assume that technical conditions (e.g. Novikov) hold when appropriate

# Martingale Representation Theorems

## Theorem (Martingale Representation)

*Under the assumption of a Brownian Filtration  
(and some technical conditions, assumed satisfied):*

- 1 A martingale  $(M_t)$  has a representation as a stochastic integral:

$$dM_t = \sigma_t^M dW_t$$

*for some adapted, square-integrable process  $(\sigma_t^M)$*

- 2 Furthermore, if  $(M_t)$  is positive (almost surely), it has a representation in terms of a generalized Geometric Brownian Motion:

$$dM_t/M_t = \sigma_t^M dW_t$$

*for some adapted, square-integrable process  $(\sigma_t^M)$*

# Trading Strategies and Arbitrage

## Definition (Trading Strategy)

A trading strategy is a  $n$ -dimensional, adapted and square-integrable process  $(\delta_t) \in \mathbb{R}^n$ , representing the holdings of the  $n$  assets at time  $t$

## Definition (PnL)

$$PnL(0, T; \delta) = \int_0^T \delta'_t dS_t \in \mathbb{R}$$

## Definition (Arbitrage)

An arbitrage is a trading strategy  $\delta$  such that:

- 1  $PnL(0, T; \delta) \geq 0$  with probability 1
- 2  $PnL(0, T; \delta) > 0$  with probability  $> 0$

# Complete Markets

## Notion and formal definition

- We are used to working with arbitrage-free markets, another important notion is that of *complete* markets
- Completeness conveys the idea that randomness is tradable
- The formal definition of a complete market is that all (square-integrable) variables are replicated by a combination of cash and a trading strategy:  
*For any variable  $X_T$  in  $L^2$ , it exists a deterministic  $\alpha_0 \in \mathbb{R}$  and an adapted, square-integrable process  $(\delta_t) \in \mathbb{R}^p$  such that:*

$$X_T = \alpha_0 + \int_0^T \delta'_t dS_t$$

- We find this definition somewhat circular: the notion of replication is best formalized under the equivalent martingale measure (EMM), we haven't introduced it yet, and we need complete markets to guarantee its unicity
- We will, instead, put together another definition, which intuitively expresses the idea of completeness, and check that our definition is equivalent

# Complete Markets

## Heuristic Approach

- The idea that randomness is tradable *could* be formalized by:

$$\forall t > 0, W_t = f(S_t, t) \text{ for some deterministic function } f$$

- This is far too restrictive: not even satisfied in a local volatility model (although it is satisfied under more restrictive Bachelier or Black and Scholes)
- It is not even generally the case that  $S_t = f(W_t, t)$
- What is always true is  $dS_t = f_t(dW_t)$ :  
price *increments*  $dS_t$  are *predictable* functions of *innovations*  $dW_t$ :

$$dS_t = \mu_t dt + \sigma_t dW_t$$

where *predictable* means: deterministic (i.e. measurable) as of time  $t$ , since all terms in the SDE other than  $dS_t$  and  $dW_t$  are  $t$ -measurable

- Hence:  $S_t = f(\{dW_s, 0 \leq s < t\}) = g(\{W_s, 0 \leq s \leq t\})$
- More generally, since our filtration is Brownian, randomness is encapsulated in Brownian paths, such that all  $t$ -measurable variables  $X_t$  are deterministic functionals of the Brownian path:  $X_t = f(\{W_s, 0 \leq s \leq t\})$

# Complete Markets

## Heuristic Approach

- Therefore, completeness must really mean: *random innovations are tradable*
- In other terms, the SDE is, in some sense, *invertible*:

$$\forall t > 0, \exists \alpha_t, \beta_t : dW_t = \alpha_t dt + \beta_t dS_t$$

- Assuming this is the case, it follows that any  $t$ -measurable variable  $X_t$  is a functional of the tradable increments:

$$X_t = f(\{W_s, 0 \leq s \leq t\}) = g(\{dW_s, 0 \leq s < t\}) = h(\{dS_s, 0 \leq s < t\})$$

so our "inverse SDE" implies the formal definition, at least in a heuristic sense

- Conversely, the formal definition implies for all  $t$ -measurable variables  $X_t$ :

$$X_t = h(\{S_s, 0 \leq s \leq t\}) = g(\{dS_s, 0 \leq s < t\})$$

pick  $X_{t+\epsilon} = W_{t+\epsilon} - W_t \rightarrow dW_t$ , it must be a predictable function of  $dS_t$

- By Ito's lemma, this function must be of the form  $dW_t = \alpha_t dt + \beta_t dS_t$

# Complete Markets

## Definition (Complete Markets)

- Markets are complete when random innovations are predictable functions of tradable increments: it exists two adapted, square-integrable processes  $(\alpha_t) \in \mathbb{R}^p$  and  $(\beta_t) \in \mathbb{R}^{p \times n}$  such that

$$dW_t = \alpha_t dt + \beta_t dS_t$$

- This is the same as: all variables are functionals of the price series  $(S_t)$

## Example (Complete and Incomplete Markets)

- In dimension 1, when  $n = p = 1$ , markets are complete as long as  $\sigma_t > 0$ :

$$dS_t = \mu_t dt + \sigma_t dW_t \iff dW_t = -\frac{\mu_t}{\sigma_t} dt + \frac{1}{\sigma_t} dS_t$$

- If  $n = p$ , markets are complete if and only if  $\sigma_t \in \mathbb{R}^{n \times n}$  is *invertible*
- With more Brownians than assets,  $n < p$ , markets are *incomplete*
  - You cannot deduce 4 independent innovations from 3 given increments!
  - e.g. stochastic volatility models are incomplete: volatility is not tradable

# Complete Markets

## Definition (Equivalent definition)

- Markets are complete if and only if  $\forall t \in [0, T] : \text{rank}(\sigma_t) = p$
- In other terms: the  $p$  columns of  $\sigma_t$  are linearly independent
- Proof:

$$\begin{aligned} dS_t = \mu_t dt + \sigma_t dW_t &\iff \sigma'_t dS_t = \sigma'_t \mu_t + (\sigma'_t \sigma_t) dW_t \\ &\iff (\sigma'_t \sigma_t) dW_t = \sigma'_t dS_t - \sigma'_t \mu_t \end{aligned}$$

- This equation has a unique solution if and only if  $(\sigma'_t \sigma_t)$  is invertible
- If and only if  $\text{rank}(\sigma_t) = p$
- In this case:  $dW_t = -\sigma'_t \mu_t + (\sigma'_t \sigma_t)^{-1} \sigma'_t dS_t$
- Implies  $p \leq n$  and reduces to  $\sigma_t > 0$  when  $n = p = 1$
- Means that the columns of  $\sigma_t$  are linearly independent at all times, with probability 1

# Derivatives Contracts

## Definition (Derivatives Contract)

A Derivatives contract is specified by a scalar, square-integrable payoff  $\pi_T$ : an arbitrary random variable known at maturity  $T$ , as indicated by the subscript

## Remark

- Under our assumption of a Brownian filtration,  $\pi_T$  is a deterministic functional of  $\{W_t, 0 \leq t \leq T\}$
- In complete markets,  $\pi_T$  is also a deterministic functional of  $\{S_t, 0 \leq t \leq T\}$
- This definition therefore includes all European options, where  $\pi_T = g(S_T)$ , including calls when  $g(x) = (x - K)^+$ , and also, all kinds of *path-dependent* options, where the payoff depends on everything that happens between inception  $t = 0$  and maturity  $T$
- This is exactly how lawyers write term sheets: anticipating every possibility, and defining corresponding cash-flows: they may not know this, but they are really instantiating functionals of paths in the appropriate universe

# Girsanov's Theorem

# Girsanov's Theorem

Girsanov's theorem will be our workhorse for this lesson

- A centerpiece of Modern Mathematical Finance, providing a clean proof of the foundational theorems
- Also a pillar of the Derivatives industry, practitioners use it constantly, for example, to:
  - ▶ Find closed-form formulas for quanto options or constant-maturity swaps
  - ▶ Compute impact of collateral on prices
  - ▶ Implement importance sampling in Monte-Carlo simulations
  - ▶ Among many others
- Students (and professionals) often often find it hard to understand or apply it
- This review demystifies the theorem and clarifies its meaning and application
- It does necessarily provides the most general and rigorous proofs

# Equivalent Probability Measures

## Definition (Equivalent Measures)

Two probability measures  $P$  and  $Q$  on  $\Omega$  are equivalent when they have the same certain and impossible events:

$$\forall A \subset \Omega, P(A) = 1 \iff Q(A) = 1 \text{ and } P(A) = 0 \iff Q(A) = 0$$

## Example (Equivalent measures redistribute mass but not support)

- Rig a dice so face 6 is more likely: equivalent measure – but if face is 1 unattainable, or a 7th face is added: not an equivalent measure
- More to the point, the quadratic variation of a Brownian Motion is  $(t)$  with probability 1 – so it must be the case under all equivalent measures
- The only processes whose QV is  $(t)$  a.s. are drifted Brownian motions  $dW_t + \mu_t dt$ , hence, Brownian Motions pick-up a drift by change of measure, but are otherwise unchanged: this is the essence of Girsanov's theorem
- Recall the definition of an arbitrage:  $P(PnL < 0) = 0$  and  $P(PnL > 0) > 0$ ; those 2 properties carry over across equivalent measures: an arbitrage remains an arbitrage under all equivalent measures!

# Radon-Nikodym Theorem

## Theorem (Radon-Nikodym)

- If  $P$  and  $Q$  are equivalent measures, then it exists a variable  $L$  such that for any other variable  $X$ :

$$E^Q[X] = E^P[LX]$$

- $L$  is called the Radon-Nikodym derivative of  $Q$  wrt  $P$  and denoted  $dQ/dP$
- $L > 0$  a.s. and  $E^P[L] = 1$

## Remark

- This is evident when the measures have densities  $\phi_P$  and  $\phi_Q$  – in this case:

$$\int_{\Omega^*} X(\omega) \phi_Q(\omega) d\omega = \int_{\Omega^*} X(\omega) \frac{\phi_Q(\omega)}{\phi_P(\omega)} \phi_P(\omega) d\omega = \int_{\Omega^*} X(\omega) L(\omega) \phi_P(\omega) d\omega$$

where  $\Omega^*$  is the common support of the two densities, which must exist by equivalence, and  $L = \phi_Q/\phi_P$  is simply the ratio of densities

- The power of the theorem is the generalization to arbitrary measures, including those without a density

# Likelihood Process

## Definition (Likelihood Process)

- Recall in our framework, all variables are revealed on or before horizon  $T$
- Hence, we can denote  $L_T$  the Radon-Nikodym derivative  $dQ/dP$
- Define  $L_t = E_t^P[L_T]$  the likelihood process
- $(L_t)$  is a positive martingale with unit expectation under  $P$ , hence it has a lognormal martingale representation of the form:

$$dL_t/L_t = \lambda'_t dW_t^P$$

for some  $p$ -dimensional, adapted process  $(\lambda_t)$  called the likelihood kernel

## Proposition (proven in optional exercises)

If  $P$  and  $Q$  are equivalent measures and  $dP/dQ = L_T$ :

- For any  $t$ -measurable variable  $X_t$ :  $E^Q[X_t] = E^P[L_T X_t] = E^P[L_t X_t]$
- $Q_t$  and  $P_t$  are also equivalent, and:  
 $dQ_t/dP_t = L_t/L_t$ , in other terms:  $E_t^Q[X_T] = E_t^P[L_T X_T]/L_t$

# Exponential Martingales and Kernel Processes

## Notations

- Positive Brownian martingales  $(M_t) \in \mathbb{R}$  with unit expectation have an *exponential martingale representation* of the form:

$$M_0 = 1 \text{ and } \frac{dM_t}{M_t} = \sigma_t^{M'} dW_t \iff M_t = \exp\left(-\int_0^t \frac{\|\sigma_t^M\|^2}{2} dt + \int_0^t \sigma_t^{M'} dW_t\right)$$

where  $(\sigma_t^M) \in \mathbb{R}^p$  is an adapted, squared-integrable process

- In order to avoid spelling-out the integral equation many times, we denote:

$$M = \zeta(\sigma^M; P)$$

where the second argument specifies the measure where  $(M_t)$  is a martingale

- We often simply call  $\sigma^M$  the volatility of  $(M_t)$ , omitting "lognormal"
- Conversely, when  $M_t = E_t^P[X_T]$  for some positive, squared-integrable variable  $X_T$  with unit expectation, we denote its *kernel* with the  $\chi$  Greek letter:

$$\sigma^M = \chi(X; P) \iff (E_t[X_T]) = \zeta(\sigma^M; P)$$

# Girsanov's Theorem

We are now ready to state Girsanov's theorem:

## Theorem (Girsanov)

$(W_t^P)$  is a  $p$ -dimensional Brownian Motion under  $P$  if and only if  $W_t^Q = W_t^P - \int_0^t \lambda_s ds$  is a  $p$ -dimensional Brownian Motion under the equivalent measure  $Q$ , where  $\lambda = \chi(dQ/dP; P)$

## Demystification

- In other words, the BM  $(W_t^P)$  picks-up a drift by equivalent change of measure  $P \rightarrow Q$ , and the acquired drift is precisely the kernel of  $dQ/dP$
- The theorem is also applicable *the other way around*: if you want your BM to pick-up a drift  $(\mu_t)$ , perform an equivalent change of measure to  $Q$  such that  $dQ/dP = \zeta(\mu; P)$

# Girsanov's Theorem

## Heuristic Proof.

- $(W_t^P) \in \mathbb{R}^p$  has quadratic *co-variation*  $t \cdot I_p$  almost surely, under  $P$  and all equivalent measures
- By Levy's characterisation theorem,  $(W_t^P)$  must be a drifted BM under all equivalent measures
- All that remains is to compute the drift under  $Q$  – Heuristically:

$$\begin{aligned} E_t^Q[dW_t^P] &= E_t^P[L_T dW_t^P]/L_t && \text{(prop 2)} \\ &= E_t^P[L_{t+dt} dW_t^P]/L_t && \text{(prop 1)} \\ &= E_t^P[(L_t + dL_t) dW_t^P]/L_t && = E_t^P[(1 + dL_t/L_t) dW_t^P] \\ &= E_t^P[dW_t^P] + E_t^P[(dL_t/L_t) dW_t^P] && = 0 + E_t^P[(\lambda_t' dW_t^P) dW_t^P] \\ &= E_t^P[dW_t^P W_t^{P'} \lambda_t] && = E_t^P[I_p \lambda_t] \\ &= \lambda_t dt \end{aligned}$$



# Girsanov's Theorem

## Corollary (Girsanov's Theorem: More Practical Statement)

Suppose  $(X_t)$  is Brownian semi-martingale and  $(Y_t)$  is a positive martingale under some measure  $P$ . Then,  $(X_t)$  picks-up an additional drift  $(\mu_t^{P \rightarrow Q})$  under the measure  $Q$  defined by  $dQ/dP = Y_T/Y_0$ , given by the expression:

$$\mu_t^{X:P \rightarrow Q} = dX_t \cdot d \log Y_t / dt$$

## Proof.

Denote  $\sigma^Y = \chi(Y_T/Y_0; P)$ ; by Girsanov,  $dW_t^Q = dW_t^P - \sigma_t^Y dt$  is a standard BM under  $Q$ , and:

$$\begin{aligned} dX_t &= \mu_t^X dt + \sigma_t^{X'} dW_t^P \\ &= \mu_t^X dt + \sigma_t^{X'} (dW_t^Q + \sigma_t^Y dt) \\ &= \mu_t^X dt + \sigma_t^{X'} dW_t^Q + \sigma_t^{X'} \sigma_t^Y dt \\ &= \mu_t^X dt + \sigma_t^{X'} dW_t^Q + dX_t \cdot d \log Y_t \end{aligned}$$



# Girsanov's Theorem

## Example (Effortlessly derive Black and Scholes' formula)

- Recall:  $d \log S_t = -\frac{\sigma^2}{2} dt + \sigma dW_t^Q$ , and:
- $C_t = E_t^Q[(S_T - K)^+] = E_t^Q[(S_T - K)\mathbb{1}_{S_T > K}] = E_t^Q[S_T \mathbb{1}_{S_T > K}] - Q(S_T > K)$

$$\begin{aligned} Q_t(S_T > K) &= Q_t(\log S_T > \log K) \\ &= Q_t(\log S_t - \frac{\sigma^2}{2}(T - t) + \sigma\sqrt{T - t}N > \log K) \\ &= N(d_2) \end{aligned}$$

- $E_t^Q[S_T \mathbb{1}_{S_T > K}] = S_t E_t^Q[S_T/S_t \mathbb{1}_{S_T > K}] = Q_t^S(S_T > K)$  where  $\frac{dQ^S}{dQ} = \frac{S_T}{S_0}$
- $(\log S_t)$  picks-up additional drift under  $Q^S$ , equal to  $(d \log S_t)^2/dt = \sigma^2$
- Hence:  $d \log S_t = +\frac{\sigma^2}{2} dt + \sigma dW_t^{Q^S}$
- $Q_t^S(S_T > K) = Q^S(\log S_t + \frac{\sigma^2}{2}(T - t) + \sigma\sqrt{T - t}N > \log K) = N(d_1)$

# Girsanov's Theorem

## Example (Exchange Option under Black and Scholes)

- $(X_t)$  and  $(Y_t)$  are the prices of two securities, assumed lognormal with volatilities  $\sigma_X$  and  $\sigma_Y$ , and correlation  $\rho$
- $(X_t)$  and  $(Y_t)$  are martingales under the risk-neutral measure  $Q$
- What is the price of the option to exchange one unit of  $X$  for  $K$  units of  $Y$ ?
- $C_t = E_t^Q[(X_T - KY_T)^+] = E_t^Q[Y_T(Z_T - K)^+]$  where we defined  $Z_t = X_t/Y_t$
- $C_t = Y_t E_t^{Q_Y}[(Z_T - K)^+]$  where  $dQ^Y/dQ = Y_T/Y_0$
- By Ito:  $dZ_t/Z_t = (\sigma_Y^2 - \rho\sigma_X\sigma_Y)dt + \sigma_X dW_t^X - \sigma_Y dW_t^Y$
- $dW_t^X - \sigma_Y dW_t^Y = \sigma_Z dB_t^Q$  where  $\sigma_Z = \sqrt{\sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y}$  and  $(B_t^Q)$  is a BM under  $Q$

# Girsanov's Theorem

## Example (Exchange Option under Black and Scholes)

- By Ito:  $d \log Z_t = \frac{\sigma_Y^2 - \sigma_X^2}{2} dt + \sigma_Z dB_t^Q$
- Under  $Q^Y$ ,  $(\log Z_t)$  picks-up an additional drift equal to:
$$(d \log Z_t)(d \log Y_t)/dt = \rho \sigma_X \sigma_Y - \sigma_Y^2$$
- So:  $d \log Z_t = -\frac{\sigma_X^2 + \sigma_Y^2 - 2\rho \sigma_X \sigma_Y}{2} dt + \sigma_Z dB_t^{Q^Y} = -\frac{\sigma_Z^2}{2} dt + \sigma_Z dB_t^{Q^Y}$
- Hence,  $(Z_t)$  is a martingale under  $Q^Y$ !  
we will see that this is *not* a coincidence when we study numeraires
- And the result is:

$$C_t = Y_t \cdot BS(Z_t, t; K, T; \sigma_Z) = BS(X_t, t; KY_t, T; \sqrt{\sigma_X^2 + \sigma_Y^2 - 2\rho \sigma_X \sigma_Y})$$

by homogeneity of BS:  $\alpha BS(x, t; K; T; \sigma) = BS(\alpha x, t; \alpha K; T; \sigma)$

# Fundamental Asset Pricing Theorems

# Fundamental Asset Pricing Theorems

## Theorem (First Theorem)

*Markets are arbitrage-free if and only if it exists an equivalent martingale measure (EMM, denoted  $Q$ ) where asset prices are martingales*

## Theorem (Second Theorem)

*Arbitrage-free markets are complete if and only if the EMM is unique*

## Corollary (Derivatives Pricing)

*Assuming arbitrage-free, complete markets:*

- 1 Any claim  $\Pi_T \in L^2$  is replicated by some cash and a trading strategy: it exists a  $t$ -measurable variable  $\pi_t \in \mathbb{R}$  a trading strategy  $(\Delta_t) \in \mathbb{R}^n$  such that

$$\Pi_T = \Pi_t + \int_t^T \Delta'_u dS_u$$

- 2 Derivatives prices are  $Q$ -martingales too: the price of  $\Pi_T$  is  $\Pi_t = E_t^Q[\Pi_T]$
- 3 Hedge ratios are sensitivities:  $\Delta_t$  is the gradient of  $\Pi_t$  wrt  $S_t$

# Proof of the First Theorem

## History

classic  
specific  
concrete  
easy



modern  
general  
abstract  
hard

1970s Black and Scholes PDE + Feynman-Kac

- for diffusions only
- see lesson 1

1980s Harrison-Kreps and Harrison-Pliska

- for semi-martingales
- see next

1990s Delbaen and Schachermayer

- for general price processes
- beautiful but overkill
- see original papers or Bjork (chp 12)

# First Theorem: Classic Proof

$\exists \text{EMM} \implies \text{no arbitrage.}$

- We already (kind of) proved this direction when we stated that an arbitrage remains an arbitrage under all equivalent measures
- Let us prove the contrapositive: arbitrage  $\implies$  no EMM
- Assume there is an arbitrage strategy  $(\delta_t)$  such that:

$$P[PnL(0, T; \delta) \geq 0] = 1 \text{ and } P[PnL(0, T; \delta) > 0] > 0$$

- Then it remains the case under all equivalent measures  $Q$ :

$$Q[PnL(0, T; \delta) \geq 0] = 1 \text{ and } Q[PnL(0, T; \delta) > 0] > 0$$

- Hence,  $E^Q[PnL(0, T; \delta)] = E^Q[\int_0^T \delta'_t dS_t] > 0$
- Therefore,  $(S_t)$  cannot be a  $Q$ -martingale



# First Theorem: Classic Proof

no arbitrage  $\implies \exists EMM$ : one-dimensional case.

Single-dimensional case is almost trivial: markets are arbitrage-free (and complete, as we have seen) as long as  $(\sigma_t) \in \mathbb{R}$  is positive

- $dS_t = \mu_t dt + \sigma_t dW_t^P = \sigma_t(dW_t^P + \frac{\mu_t}{\sigma_t} dt) = \sigma_t dW_t^Q$
- Where  $dQ/dP = \zeta(-\frac{\mu}{\sigma}; P)_T$
- $(S_t)$  is a martingale under  $Q$ , hence,  $Q$  is an EMM
- The existence of an EMM guarantees no arbitrage by the previous argument



# First Theorem: Classic Proof

no arbitrage  $\implies \exists EMM$ : general case, intuition.

- The multi-dimensional case is fundamentally different, because then, strategies can hedge the risk of some assets with other assets
- For instance, consider two assets driven by the same BM:

$$dS_t^1 = \mu_t^1 dt + \sigma_t^1 dW_t \text{ and } dS_t^2 = \mu_t^2 dt + \sigma_t^2 dW_t$$

- If for some  $t$ ,  $\mu_t^1/\sigma_t^1 \neq \mu_t^2/\sigma_t^2$  (say  $\mu_t^1/\sigma_t^1 > \mu_t^2/\sigma_t^2$ ) we have an arbitrage:
- Buy 1 unit of the first asset, hedge with  $\sigma_1/\sigma_2$  units of the second:

$$dPnL = \mu_1 - \frac{\sigma_1}{\sigma_2} \mu_2 > 0$$

- Therefore, the drift vector  $(\mu_t)$  and volatility matrix  $(\sigma_t)$  cannot be arbitrary: they must be restricted in order to prevent this kind of arbitrage, in our example: it must be that  $\forall t : \mu_t^1/\sigma_t^1 = \mu_t^2/\sigma_t^2$
- Remark: these *deterministic* arbitrages are not the only ones, there are *random* arbitrages where  $PnL \geq 0$  a.s. and  $P(PnL > 0) > 0$ , but absence of arbitrage must *imply* absence of deterministic arbitrage



# First Theorem: Classic Proof

no arbitrage  $\implies \exists EMM$ : general case, formal proof.

- We now derive the general condition to prevent deterministic arbitrage – consider holdings  $\delta_t \in \mathbb{R}^n$  of the  $n$  assets at time  $t$ , the (differential) PnL is:

$$dPnL = \delta'_t dS_t = \delta'_t \mu_t dt + \delta'_t \sigma_t dW_t$$

- Risk-less PnL must be 0:

$$\delta'_t \sigma_t = 0 \iff \sigma'_t \delta_t = 0 \implies \delta'_t \mu_t = 0$$

- In other terms, it must be that:  $\text{null}(\sigma'_t) \perp \mu_t$
- By a fundamental result of Linear Algebra:  
*the orthogonal complement of the nullspace of a matrix  $M$  is the range of  $M'$*
- Hence, the condition above translates to:  $\mu_t \in \text{range}(\sigma_t)$
- In other terms, it exists  $\lambda_t \in \mathbb{R}^p$  such that  $\mu_t = \sigma_t \lambda_t$



# First Theorem: Classic Proof

no arbitrage  $\implies \exists EMM$ : general case, formal proof.

- The one-dimensional case was resolved by defining a process  $\lambda_t$  with the equation:

$$\mu_t = \lambda_t \sigma_t \iff \lambda_t = \mu_t / \sigma_t$$

which trivially exists, and is unique, as long as  $\sigma_t > 0$

- In the general case, it is not given that a solution to this equation exists
- What we have just seen is that its existence precisely corresponds to absence of deterministic arbitrage
- The proof completes itself, as it follows that:

$$dS_t = \sigma_t (dW_t^P + \lambda_t dt) = \sigma_t dW_t^Q$$

where  $dQ/dP = \zeta(-\lambda; P)_T$



# Proof of Second Theorem

## Second Theorem.

- The argument for the first theorem revolved around the equation:

$$\mu_t = \sigma_t X_t$$

- $Q$  is an EMM if and only if  $dQ/dP = \zeta(-x; P)_T$  where  $(x_t)$  is a solution
- Absent deterministic arbitrage, we showed that the equation must have a solution, proving existence
- The EMM is unique if and only if the solution  $(x_t)$  is unique
- If and only if the rank of  $\sigma_t$  is  $p$ , i.e. its  $p$  columns are linearly independent (which, in particular, implies that  $n \geq p$ )
- This is a definition of complete markets, completing the proof



# Proof of Corollary

## Derivatives Pricing.

- Assuming complete, arbitrage-free markets, it exists a unique EMM  $Q$  defined by:

$$dS_t = \sigma_t dW_t^Q$$

- Notice, the volatility matrix process is still  $(\sigma_t)$  under the EMM:
  - This is an explicit result in the proof of the First Theorem
  - And may be re-demonstrated by a quadratic variation argument:

$$QV(0, t) = \int_0^t \text{tr}(\sigma'_t \sigma_t) dt \text{ a.s.}$$

so it must be the case under all equivalent measures

- A claim is defined as an arbitrary, square-integrable variable  $\Pi_T \in \mathbb{R}$ ; *define*:

$$\Pi_t = E_t^Q[\Pi_T]$$



# Proof of Corollary

## Derivatives Pricing.

- Then  $(\Pi_t)$  is a Q-martingale, hence, by the martingale representation theorem (recall the filtration is Brownian by assumption) it exists an adapted, squared-integrable process  $(\beta_t) \in \mathbb{R}^p$  such that:

$$\Pi_T = \Pi_t + \int_t^T \beta'_u dW_u^Q$$

- Here, we use market completeness:  $dW_u^Q = \dots du + \gamma_u dS_u = \gamma_u dS_u$  since  $(S_t)$  and  $(W_t^Q)$  are both martingales under Q
- So, finally:  $\Pi_T = \Pi_t + \int_t^T \beta'_u \gamma_u dS_u = \Pi_t + \int_t^T \delta'_u dS_u$  where we defined  $\delta_t = \gamma'_t \beta_t \in \mathbb{R}^n$
- $\Pi_T$  is replicated with cash  $\Pi_t$  and the PnL of a trading strategy  $(\delta_t)$ : it follows that  $\Pi_t$  must be its price at  $t$ , proving that Derivatives prices are martingales



# Proof of Corollary

## Derivatives Pricing.

- It remains to prove that  $\delta_t$  is the gradient  $\partial \Pi_t / \partial S_t$
- This is delicate in general, since:
  - ▶ path-dependent option prices are path functionals:

$$\Pi_t = f(\{S_s, s \in [0, t]\})$$

- ▶ the volatility ( $\sigma_t$ ) of semi-martingales may be path-dependent too:

$$\sigma_t = f(\{S_s, s \in [0, t]\})$$

and classic stochastic calculus does not define derivatives of such things

- In the special case where the model is a diffusion and the option is European:

$$\sigma_t = \sigma(S_t, t) \text{ and } \Pi_T = \Pi(S_T)$$

we know by Markov property that:

$$\Pi_t = \Pi(S_t, t)$$



# Proof of Corollary

## Derivatives Pricing.

- In this special case, by Ito's lemma:

$$d\Pi_t = \left(\frac{\partial \Pi_t}{\partial S_t}\right)' dS_t + \dots dt = \left(\frac{\partial \Pi_t}{\partial S_t}\right)' dS_t$$

since  $(\Pi_t)$  and  $(S_t)$  are both Q-martingales

- Hence:

$$\Pi_T - \Pi_t = \int_t^T \delta'_u dS_u = \int_t^T \left(\frac{\partial \Pi_u}{\partial S_u}\right)' dS_u$$

- Proving that  $\delta_t = \partial \Pi_t / \partial S_t$  by unicity of stochastic integral representation
- In the general case of a path-dependent option and/or SDE, Dupire calculus (2008) rigorously defines calculus for path functionals and demonstrates that it is still the case



# Numeraires

# Numeraire

- Nowhere in our proofs did we mandate a *unit* for asset prices ( $S_t$ ), Derivatives prices ( $\Pi_t$ ) (including payoff  $\Pi_T$ ) or wealth in general, expressed with stochastic integrals
- It follows that the theorems are *numeraire-agnostic*: they must be true irrespective of choice of unit
- Numeraire could be US dollars, Euros, AAPL shares, the price per sqf in Manhattan, or whatever: *and the Theorems must still hold*
- Evidently:
  - ▶ Units must be *consistent*: you can't mix USD and EUR in your derivations
  - ▶ The numeraire must be *positive* or prices are not well-defined
- Besides (in complete, arbitrage-free markets) prices are unique: it must be that prices in EUR are prices in USD over EUR/USD
- And sensitivities  $\partial\Pi/\partial S$  are dimensionless: they must be the same regardless of numeraire

# Re-expression of the Fundamental Theorems

## We therefore generalize the Theorems:

Markets are complete and arbitrage-free if and only if for any valid choice of numeraire  $(N_t) \in \mathbb{R}_+$ , it exists a unique EMM  $Q^N$  such that asset prices, both underlying or derivative, **expressed in terms of this numeraire**, are  $Q^N$ -martingales:

$$d\left(\frac{S_t}{N_t}\right) = \sigma_t^{S/N} dW_t^N \text{ and } d\left(\frac{\Pi_t}{N_t}\right) = \sigma_t^{\Pi/N'} dW_t^N$$

for some adapted, square integrable processes  $(\sigma_T^{S/N}) \in \mathbb{R}^{n \times p}$  and  $(\sigma_T^{\Pi/N'}) \in \mathbb{R}^p$

## Replication results hold, with sensitivities independent of numeraire:

$$\frac{\partial \frac{\Pi_t}{N_t^1}}{\partial \frac{S_t}{N_t^1}} = \frac{\partial \frac{\Pi_t}{N_t^2}}{\partial \frac{S_t}{N_t^2}} = \frac{\partial \Pi_t}{\partial S_t}$$

# Example

## Example (SP500 call in USD and EUR)

- Assume no rates or dividends and denote  $(S_t)$  the SP500 quoted in USD, and  $(F_t)$  the EUR/USD rate, i.e. the price of 1 EUR, also quoted in USD
- Assume  $(S_t)$  and  $(F_t)$  are lognormal processes, with respective volatilities  $\sigma_X$  and  $\sigma_F$ , and correlation  $\rho$
- We want to price the call of some strike  $K$  *expressed in USD* and maturity  $T$ , with USD or EUR as the numeraire
- With USD as numeraire, we know the result:  $C_t^{USD} = BS(S_t, t; K, T; \sigma_X)$
- Evidently, it must be that:  $C_t^{EUR} = C_t^{USD} / F_t$
- Let us check how this is the case, using EUR as numeraire

# Example

## Example (SP500 call in USD and EUR)

- Choose EUR as the numeraire, and work under the EUR EMM  $Q^{EUR}$
- Express prices in EUR:  $Z_t = S_t/F_t$  is the price of SP500 in EUR, then:
  - ▶  $(Z_t)$  is a martingale under  $Q^{EUR}$
  - ▶ actually: a lognormal martingale with volatility:  $\sigma_Z = \sqrt{\sigma_X^2 + \sigma_F^2 - 2\rho\sigma_X\sigma_F}$
- The payoff is  $(S_T - K)^+$  USD, which is  $(Z_T - K/F_T)^+$  EUR
- The EUR price of the call is a martingale under  $Q^{EUR}$ , hence:

$$C_t^{EUR} = E^{EUR}[(Z_T - K/F_T)^+]$$

- This is the option to exchange 1 unit of  $Z$  for  $K$  units of  $1/F$   
besides,  $(1/F_t)$  is the price of 1 USD in EUR, a martingale under  $Q^{EUR}$

# Example

## Example (SP500 call in USD and EUR)

- Reuse the formula we found earlier for the exchange option, applied to underlying series  $(Z_t)$  and  $(1/F_t)$ , both prices in EUR, both  $Q^{EUR}$  martingales:

$$\begin{aligned}C_t^{EUR} &= BS(Z_t, t; K(1/F_t), T; \sigma_{ZF}) &= BS(X_t/F_t, t; K/F_t, T; \sigma_X) \\ &= (1/F_t)BS(X_t, t; K, T; \sigma_X) &= C_t^{USD}/F_t\end{aligned}$$

- Notice:

We translated all prices in Euros, resulting in different price series with different volatilities, and a different payoff, yet, final results are consistent, as promised by the abstract proof!

# Bank Account Numeraire

- Before we introduced numeraires, we *implicitly* used some currency, such as USD or EUR
- In the previous example, we used currencies explicitly
- We use currencies numeraires in our daily lives, but they are *not* meaningful from an economic standpoint
- Cash is deposited in bank accounts, and earns interest:  
the value of a bank account, conventionally denoted  $(\beta_t)$ , is therefore:

$$\beta_t = \exp\left(\int_0^t r_s ds\right)$$

where we set  $\beta_0 = 1$  and  $(r_t)$  is the short rate series

# Bank Account Numeraire

- The bank account  $(\beta_t)$ , not cash, is the natural numeraire
- The associated EMM is called *risk-neutral measure* and denoted  $Q = Q^\beta$
- Under this EMM:
  - ▶ Asset prices *in terms of*  $(\beta_t)$  are martingales:

$$d\left(\frac{S_t}{\beta_t}\right) = \sigma_t dW_t^\beta \text{ hence } dS_t = r_t S_t dt + \sigma_t dW_t^\beta$$

the drift of asset prices is the short rate, which justifies the name "risk-neutral"

- ▶ Derivatives prices are expected *discounted* payoffs:

$$\frac{\Pi_t}{\beta_t} = E_t^\beta\left[\frac{\Pi_T}{\beta_T}\right] \text{ hence } \Pi_t = E_t^\beta\left[\frac{\beta_t}{\beta_T}\Pi_T\right] = E_t^\beta\left[\exp\left(-\int_t^T r_u du\right)\Pi_T\right]$$

- As promised, we re-introduced rates in a natural and rigorous manner

# Change of Numeraire

We said earlier that practitioners use Girsanov's theorem all the time:

*We lied*

What we do use all the time is the change of numeraire formula below, which encapsulates Girsanov's theorem and enables effortless changes of measure

## Theorem (Change of Numeraire Formula (El Karoui and al. 1995))

*Consider two numeraires  $(N_t^1)$  and  $(N_t^2)$  and their associated EMMs  $Q^1$  and  $Q^2$ . Moving from  $Q^1$  to  $Q^2$  causes all semi-martingales  $(X_t)$  to pick-up an additional drift equal to:*

$$\mu_t^{X:Q^1 \rightarrow Q^2} = dX_t \cdot (d \log N_t^2 - d \log N_t^1)$$

# Change of Numeraire

## Proof (Change of Numeraire).

- We can price an arbitrary claim  $\Pi_T$  under numeraire 1 or 2:

$$\frac{\Pi_0}{N_0^1} = E^1\left[\frac{\Pi_T}{N_T^1}\right] \text{ and } \frac{\Pi_0}{N_0^2} = E^2\left[\frac{\Pi_T}{N_T^2}\right]$$

- Hence:

$$E^1\left[\frac{N_0^1}{N_T^1} \Pi_T\right] = E^2\left[\frac{N_0^2}{N_T^2} \Pi_T\right]$$

- Define  $Y = \frac{N_0^2}{N_T^2} \Pi_T$ :

since  $\Pi_T$  is arbitrary in  $L^2$ , so is  $Y$  and:

$$E^2[Y] = E^1\left[\frac{N_0^1}{N_T^1} / \frac{N_0^2}{N_T^2} Y\right] = E^1\left[\frac{N_T^2 N_0^1}{N_T^1 N_0^2} Y\right]$$

- Hence:  $dQ^2/dQ^1 = \frac{N_T^2 N_0^1}{N_T^1 N_0^2} = Z_T$ ;  $E^1[Z_T] = 1$ ; denote:  $Z_t = E_t^1[Z_T] = \frac{N_t^2 N_0^1}{N_t^1 N_0^2}$
- By the corollary to Girsanov's theorem,  $(X_t)$  picks-up a drift:

$$\mu_t^{X:Q^1 \rightarrow Q^2} = dX_t \cdot d \log Z_t / dt = dX_t \cdot (d \log N_t^2 - d \log N_t^1) / dt$$



# Change of Numeraire

## Example (Revisiting the Exchange Option)

- We found an analytic formula for the exchange option  $(X_T - KY_T)^+$  in Black and Scholes earlier in this lesson
- It certainly was not effortless: two applications of Ito and one of Girsanov
- We re-derive the formula in a few lines with the change of numeraire formula
- Using  $(Y_t)$  as numeraire under its EMM  $Q^Y$ :

$$C_t = Y_t E^Y \left[ \frac{(X_T - KY_T)^+}{Y_T} \right] = Y_t E^Y \left[ \left( \frac{X_T}{Y_T} - K \right)^+ \right]$$

- $(X_t/Y_t)$  is a lognormal martingale under  $Q_Y$ , with variance

$$\sigma_{X/Y}^2 = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y$$

- Hence:  $C_t = Y_t \cdot BS(X_t/Y_t, t; K; T; \sigma_{X/Y}) = BS(X_t, t; KY_t, T; \sigma_{X/Y})$   
by homogeneity of BS